17. Write a short note on 'electrostatic shielding'.
18. What is polarisation?
19. What is dielectric strength?
20. Define 'capacitance'. Give its unit.
21. What is corona discharge?

III Long Answer questions

1. Discuss the basic properties of electric charges.
2. Explain in detail Coulomb's law and its various aspects.
3. Define 'electric field' and discuss its various aspects.
4. Calculate the electric field due to a dipole on its axial line and equatorial plane.
5. Derive an expression for the torque experienced by a dipole due to a uniform electric field.
6. Derive an expression for electrostatic potential due to a point charge.
7. Derive an expression for electrostatic potential due to an electric dipole.
8. Obtain an expression for potential energy due to a collection of three point charges which are separated by finite distances.
9. Derive an expression for electrostatic potential energy of the dipole in a uniform electric field.
10. Obtain Gauss law from Coulomb's law.
11. Obtain the expression for electric field due to an infinitely long charged wire.
12. Obtain the expression for electric field due to an charged infinite plane sheet.
13. Obtain the expression for electric field due to an uniformly charged spherical shell.
14. Discuss the various properties of conductors in electrostatic equilibrium.
15. Explain the process of electrostatic induction.
16. Explain dielectrics in detail and how an electric field is induced inside a dielectric.
17. Obtain the expression for capacitance for a parallel plate capacitor.
18. Obtain the expression for energy stored in the parallel plate capacitor.
19. Explain in detail the effect of a dielectric placed in a parallel plate capacitor.
20. Derive the expression for resultant capacitance, when capacitors are connected in series and in parallel.
21. Explain in detail how charges are distributed in a conductor, and the principle behind the lightning conductor.
22. Explain in detail the construction and working of a Van de Graaff generator.

Exercises

1. When two objects are rubbed with each other, approximately a charge of 50 nC can be produced in each object. Calculate the number of electrons that must be transferred to produce this charge.

   Ans: 31.25×10^{10} electrons
2. The total number of electrons in the human body is typically in the order of 10^{28}. Suppose, due to some reason, you

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